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Fuel-aware autonomous docking using RL-augmented MPC rewards for on-orbit refueling

Mahya Ramezani [a], M. Amin Alandihallaj [b,*], Barış Can Yalçın [c], Miguel Angel Olivares Mendez [c], Andreas M. Hein [b]

[a] Automation and Robotics Research Group (ARG), Interdisciplinary Centre for Security, Reliability and Trust (SnT), University of Luxembourg, Luxembourg
[b] Space Systems Engineering (SpaSys) Research Group, Interdisciplinary Centre for Security, Reliability and Trust (SnT), University of Luxembourg, Luxembourg
[c] Space Robotics (SpaceR) Research Group, Interdisciplinary Centre for Security, Reliability and Trust (SnT), University of Luxembourg, Luxembourg

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ABSTRACT

The operational lifespan of satellites is constrained by finite fuel reserves, limiting their maneuverability and mission duration. On-orbit refueling offers a transformative solution, extending satellite functionality, reducing costs, and enhancing sustainability. However, the precise execution of docking maneuvers remains a critical challenge, exacerbated by fuel sloshing effects in microgravity, which introduce unpredictable disturbances. This study proposes an integrated control framework combining Model Predictive Control (MPC) and Reinforcement Learning (RL) to ensure safe and efficient docking under these dynamic conditions. Initially, a Proximal Policy Optimization (PPO)-based RL control strategy is introduced, leveraging MPC for trajectory optimization. To further enhance adaptability in highly dynamic environments, Soft Actor-Critic (SAC) is incorporated, offering superior sample efficiency and robustness against stochastic disturbances. The proposed SAC-MPC framework effectively mitigates fuel sloshing effects by balancing computational efficiency with predictive accuracy. Experimental validation is conducted in the Zero-G Lab, emulating control scenarios with 3-DoF floating platforms, while high-fidelity numerical simulations extend the study to 6-DoF dynamics with realistic sloshing behavior modeled using OpenFOAM. Comparative results demonstrate that SAC-MPC outperforms conventional RL and MPC-based methods in docking success rate, precision, and control effort. This research establishes a robust foundation for autonomous satellite docking, contributing to the viability of on-orbit refueling missions and the future of sustainable space operations.

1. Introduction

The operational lifespan of spacecraft and satellites is inherently limited by their finite fuel reserves, which also restricts their capacity to explore distant regions of space. On-orbit satellite refueling has emerged as a transformative innovation, capable of extending satellite operations, reducing mission costs, and enhancing overall adaptability [1–3]. This approach is particularly advantageous for satellites in Geostationary Earth Orbit (GEO), which play a vital role in ensuring uninterrupted communication and weather monitoring [4].

On-orbit refueling offers significant economic and environmental benefits by reducing the frequency of satellite launches and mitigating the accumulation of space debris [5]. Furthermore, it facilitates a range of critical activities, including satellite servicing, debris removal, and the construction of advanced space infrastructure such as stations and telescopes. These advancements enhance space safety and sustainability while supporting extended human exploration [6,7].

The successful realization of on-orbit refueling hinges on the ability to perform safe and reliable rendezvous and docking operations. These operations pose substantial challenges due to the high-risk nature of close-proximity maneuvers and the severe consequences of even minor velocity miscalculations during docking. A critical concern is the dynamic behavior of fuel in a microgravity environment, where sloshing effects can lead to unpredictable and undesirable movements. Consequently, the development of a robust control strategy is imperative. Such a strategy must effectively address fuel sloshing disturbances while integrating advanced collision avoidance mechanisms to ensure operational safety and reliability.

Accurate modeling of dynamic systems is a fundamental prerequisite for effective control. In the context of fuel sloshing, significant research

* Corresponding author.
E-mail address: amin.hallaj@uni.lu (M. Amin Alandihallaj).

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has focused on understanding and characterizing the behavior of liquid propellants in space. For instance, the dynamics of spin-stabilized spacecraft with fuel sloshing were simulated by Ref. [8], while a pendulum model for fuel sloshing was introduced in Ref. [9]. Veldman et al. [10] employed computational fluid dynamics (CFD) to study liquid sloshing behavior onboard spacecraft, and Marsell et al. [11] used a similar CFD approach to model sloshing in non-spinning spacecraft. Numerical methods have also been employed to investigate the impact of fuel sloshing on spacecraft guidance, navigation, and control (GNC) systems [12].

Several studies have sought to develop mathematical models for sloshing dynamics. A nonlinear formulation of fluid sloshing dynamics onboard spacecraft was derived in Ref. [13], while stability analysis using a pendulum model was conducted in Refs. [14,15]. Comparative studies between mass-spring and pendulum models for spacecraft fuel sloshing have also been performed [16]. Parameter estimation for the mass-spring model was addressed in Ref. [17], and a Fourier–Bessel series expansion method was utilized to model spacecraft with multiple propellant tanks in Ref. [18]. Additionally, an open-source fuel sloshing model using a multi-pendulum approach was introduced in Ref. [19] to simulate various tank configurations.

The influence of fuel sloshing on satellite attitude dynamics has received considerable attention. For example, the effect of sloshing on the attitude motion of spinning spacecraft was analyzed using a spherical pendulum model [20], and the nonlinear attitude dynamics of flexible satellites were derived using a moving pulsating ball model [21]. To mitigate sloshing-induced disturbances, researchers have investigated a variety of control strategies, including fuzzy control [22–24], positive position feedback control [25], proportional-derivative (PD) control [26,27], linear quadratic regulator (LQR) methods [28–30], wave-based attitude control [31], PDE-based control [32], H∞ control [33,34], output-feedback attitude control [35], sliding mode control [36,37], model predictive control (MPC) [38,39], and Lyapunov-based feedback laws [40].

Despite extensive research on attitude dynamics, the effect of fuel sloshing on spacecraft trajectories remains relatively underexplored. For instance, a cascade nonlinear observer was designed in Ref. [41] to estimate large slosh amplitudes during spacecraft maneuvers, using a pendulum model. Jaime et al. [42] addressed thrust vector control for upper-stage rockets with fuel slosh dynamics, employing a mass-spring model and Lyapunov-based feedback control. Similarly, nonlinear feedback control laws were proposed in Ref. [43] for translational velocity and attitude control, considering sloshing modeled as a mass-spring system. Additional work has explored the use of linear matrix inequalities (LMI) for controlling multi-mass-spring sloshing models [44,45] and adaptive pole placement for spacecraft with pendulum-modeled sloshing [46]. More recently, MPC has been applied to control sloshing disturbances modeled as a spherical pendulum [47,48].

Most existing studies rely on simplified models such as single or multi-mass-spring and pendulum models, often focusing on planar motion or three degrees of freedom (DOF). However, practical applications demand control strategies encompassing all 6-DOF, particularly during docking phases where minor deviations can result in mission-critical failures. This underscores the necessity for incorporating high-fidelity CFD-based control methods to achieve precise control. Addressing this gap is the primary objective of this study.

A straightforward approach to addressing the challenges posed by fuel sloshing during docking involves predicting the movement of fuel within the satellite and mitigating its impact on the docking process. MPC is a promising candidate for this purpose due to its robust capabilities in aerospace guidance, which include forecasting future state changes and counteracting unwanted movements. MPC operates by iteratively solving an optimal control problem while continuously updating initial conditions with real-time data.

Despite its advantages, the application of MPC in scenarios involving complex nonlinear dynamics and constraints faces notable limitations [49]. Chief among these is the significant computational demand, which can restrict the frequency of guidance updates and, consequently, the system's responsiveness [50]. Additionally, the effectiveness of MPC is contingent on the availability of highly accurate models of fuel sloshing, thereby limiting its reliability in real-world operational contexts. If complex, high-fidelity models are employed to enhance prediction accuracy, the increased computational burden may render MPC unsuitable for real-time applications. This trade-off highlights the need for innovative approaches to balance model precision and computational efficiency, a challenge this study seeks to address.

In contrast to conventional control methods, Artificial Intelligence (AI), particularly Reinforcement Learning (RL), has demonstrated potential across various real-world applications, from robotics to aerospace with complex and unknown dynamics such as space robotic control [51], satellite task scheduling [52,53], satellite collision avoidance [54], satellite orbital maneuvering [55–57], formation control [58], space mission planning [59], and satellite failure prediction [60]. RL utilizes training data derived from direct interactions within simulated environments structured as Markov decision processes (MDPs). This iterative improvement process, driven by feedback from numerical rewards that evaluate the performance of the control network at each step, has been effectively employed to design robust guidance systems for diverse aerospace mission scenarios [61–64]. The exploration nature of RL inherently protects against model inaccuracies, and by concentrating the bulk of computational efforts during the pre-flight training phase, it significantly reduces the computational load during actual missions. However, one notable drawback of RL is the extensive training required to ensure that the system reaches an optimal solution. To address this, the training process can be expedited by integrating RL with a control method that already exhibits satisfactory performance, thereby enhancing training efficiency [65].

To solve this issue, this paper combines MPC and RL approaches. Firstly, we introduce a RL-based control method leveraging Proximal Policy Optimization (PPO) integrated with the principles of MPC. PPO is a simple and straightforward RL algorithm that ensures stable policy updates by constraining changes to prevent drastic fluctuations. Its effectiveness in various control applications, including robotic and aerospace systems, makes it a suitable choice for initial docking control under uncertainty. By incorporating MPC principles, we enhance PPO's predictive capabilities, allowing it to generate more structured and optimized control sequences while maintaining adaptability to environmental uncertainties. This integration provides a baseline RL-MPC framework for satellite docking operations.

However, the dynamic nature of aerospace environments, especially during docking maneuvers influenced by fuel sloshing, demands a more sophisticated approach capable of handling complex, nonlinear disturbances. Fuel sloshing introduces unpredictable forces that can affect the satellite's trajectory and attitude, making it crucial to employ a control method that efficiently adapts to these uncertainties. To address this challenge, we enhance the initial PPO-based framework by incorporating Soft Actor-Critic (SAC) alongside MPC. SAC, unlike PPO, utilizes entropy maximization to promote exploration while optimizing both the policy and value functions more efficiently. This results in improved sample efficiency and enhanced adaptability, which are essential for real-time applications in highly dynamic aerospace environments.

By transitioning from a PPO-MPC framework to a SAC-MPC approach, we develop a more resilient docking control strategy that effectively accounts for fuel sloshing disturbances. Our contribution lies in designing an integrated control methodology that leverages the predictive power of MPC and the adaptability of RL, enabling precise docking even in uncertain and dynamically changing conditions. This study not only bridges the gap between RL-based control and practical aerospace applications but also introduces a robust solution for future autonomous satellite docking missions requiring fuel-efficient and disturbance-resilient guidance strategies. The primary contributions and innovations include.

- We integrate MPC's optimal trajectory outputs directly into the RL reward function. This guides the agent toward near-optimal docking paths, significantly accelerating training and improving trajectory tracking, robustness, and adaptability.
- We employ OpenFOAM's *interFoam* solver to generate realistic, microgravity sloshing forces and torques. These CFD-derived disturbance envelopes enable MPC and RL to account for complex fuel behavior, ensuring reliable control under uncertain conditions.
- We extend a PPO-MPC baseline to a SAC-MPC framework. By using SAC's entropy-regularized learning, our approach achieves superior sample efficiency and faster convergence, effectively compensating for nonlinear slosh disturbances during critical docking phases.
- We validate the SAC-MPC strategy on 3-DoF floating platforms in the Zero-G Lab at the University of Luxembourg and in 6-DoF high-fidelity simulations.

By integrating predictive control optimization with adaptive RL methods and accurately modeling fuel sloshing dynamics, this research advances autonomous docking strategies, contributing significantly to the practical realization of efficient and sustainable on-orbit refueling operations.

The paper is organized as follows: Section 2 outlines the general dynamic model for orbital refueling missions. Section 3 presents the MPC framework, and Section 4 discusses the RL methodology and the proposed integrated framework. Section 5 presents experimental results to validate the effectiveness of the proposed control strategy in achieving precise and safe docking maneuvers, considering the dynamics of fuel sloshing. Section 6 concludes the paper with remarks on the contributions and significance of the proposed strategies for enhancing the viability of on-orbit satellite refueling missions.

2. System description

This section outlines the system dynamics and the environmental conditions central to the study. The phenomenon of fuel sloshing, characterized by the oscillatory motion of a liquid's free surface within a container, is influenced by factors such as tank geometry and external accelerations, including gravity. The primary sloshing motion can significantly alter the liquid's center of mass, inducing system-wide oscillations that impact overall dynamics. Various models, such as single or multi-mass-spring systems and pendulum-based representations, have been proposed in the literature to approximate slosh dynamics. While these models offer reasonable accuracy under terrestrial conditions, their applicability diminishes in the microgravity environment of space, where fluids exhibit non-cohesive behavior and can fragment into multiple blobs. Consequently, these models fail to fully capture the complexities of fluid behavior in space.

To address these limitations and achieve a more precise representation of sloshing dynamics in microgravity, we employ the CFD tool, OpenFOAM [66]. OpenFOAM provides robust capabilities for simulating complex fluid flow phenomena, enabling the modeling of challenges specific to space, such as fluid fragmentation and irregular motion. By leveraging OpenFOAM, we can simulate and analyze the behavior of liquids under microgravity conditions with a high degree of accuracy, thereby improving the reliability of the control strategies developed in this study.

The system configuration is illustrated in Fig. 1, depicting the mission scenario where a tanker satellite executes controlled maneuvers to dock safely with a target satellite in Low Earth Orbit (LEO). Both satellites operate within the same orbital plane. Although the system dynamics involve 6-DOF, the study emphasizes planar motion due to the constraints of the experimental setup in the Zero-G Lab of the University of Luxembourg [67], where only planar dynamics can be effectively replicated.

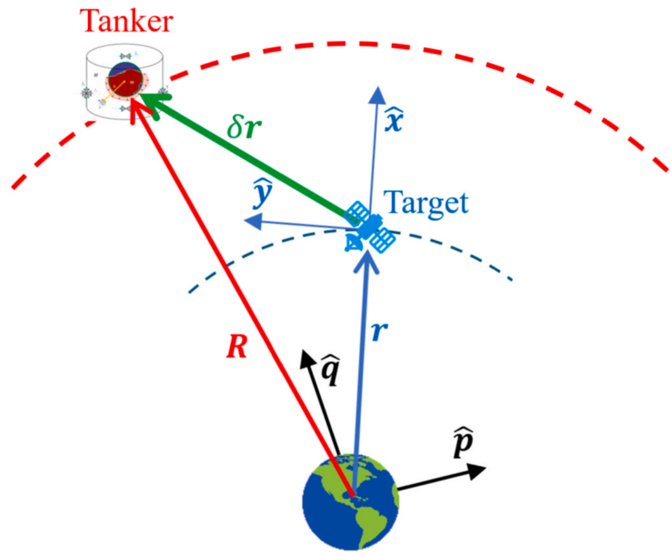

Fig. 1. The schematic diagram of the system configuration.

The Zero-G Lab at the University of Luxembourg (Fig. 2) is a critical component of the experimental framework. It replicates microgravity conditions, enabling realistic testing and validation of satellite technologies and control methodologies. By simulating reduced gravity environments, the lab provides researchers with the means to conduct experiments that closely mimic the dynamics of in-orbit satellite operations. This capability is instrumental in developing and refining control strategies for precise and safe docking maneuvers.

The lab's experimental setup supports 3-DoF motion, including both rotational and planar movements. However, since the focus of this study is specifically on a satellite refueling scenario, the performance of the 6-DoF system is studied and the 3-DoF model is utilized for lab experiments. For the purposes of these experiments, we assume that the tanker satellite has already completed the orbital maneuvers required to align its trajectory with that of the target satellite. The study then concentrates on the critical planar rendezvous and docking maneuvers.

The system comprises a rigid spacecraft, referred to as the tanker, operating within the same orbital plane as the target satellite. The tanker is equipped with a spherical fuel tank, as depicted in Fig. 3. The spacecraft's dry mass, encompassing its structural components, is denoted as M, while the fuel mass is represented as m, positioned at a distance b from the spacecraft's center of mass. To facilitate attitude

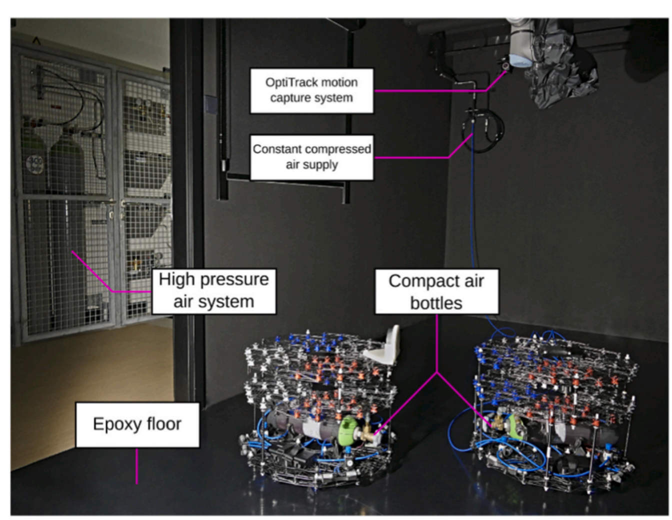

Fig. 2. Two floating platforms operating in Zero-G Lab [68].

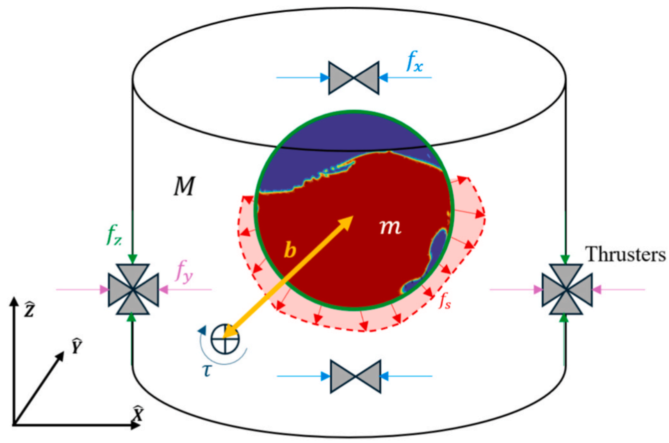

Fig. 3. The tanker model with sloshing dynamics in the spherical tank.

control and precise maneuvering, the tanker is equipped with 12 on/off thrusters. Within the experimental environment of the ZeroG Lab, the lab floor serves as an analog for the orbital plane, providing a baseline for simulating microgravity conditions. Given the experimental limitations of the lab, where only 3-DoF are available, the floating platforms are equipped with 8 thrusters.

To simplify the derivation of motion equations and experimental implementation, the following assumptions are made.

- The satellite body is modeled as rigid and non-flexible.
- System specifications are fixed throughout the study.
- The position and attitude of the target satellite are known and remain constant.
- The position, attitude, and their respective derivatives for the tanker are measurable in real time.
- Orbital disturbances, such as those induced by gravity gradients, are neglected.
- External environmental torques such as gravity gradient, drag, and magnetic disturbances are not included.

These assumptions provide a controlled framework for evaluating the performance of the proposed control strategies, ensuring the results are directly applicable to the intended scenarios of satellite rendezvous and docking. The objective is to isolate and evaluate the effects of propellant sloshing on the docking control performance.

In case the effect of sloshing is considered as a disturbance and the target satellite is in a circular orbit, the Clohessy-Wiltshire (CW) equations can describe the simplified model of relative orbital motion as [69].

$$\begin{aligned} \ddot{x} &= 3n^2x + 2n\dot{y} + \frac{F_x}{M} + d_{sx} \\ \ddot{y} &= -2n\dot{x} + \frac{F_y}{M} + d_{sy}\ddot{z} = -n^2 z + \frac{F_z}{M} + d_{sz} \end{aligned} \tag{1}$$

where n is the orbital rate of the target satellite, $\delta\boldsymbol{r} = [x \ y \ z]^{\mathrm{T}}$ represents the tanker position with respect to the target satellite expressed in the RSW coordinate frame, $\boldsymbol{F_c} = \begin{bmatrix} F_x & F_y & F_z \end{bmatrix}^{\mathrm{T}} = \boldsymbol{C}_B^R \begin{bmatrix} f_x & f_y & f_z \end{bmatrix}$ is the thruster force vector in the RSW frame, $\boldsymbol{C}_B^R$ is the body frame to RSW frame rotation matrix, and $\boldsymbol{d}_s = \begin{bmatrix} d_{sx} & d_{sy} & d_{sz} \end{bmatrix}^{\mathrm{T}} = \boldsymbol{C}_B^R \frac{\boldsymbol{f}_s}{M}$ represents the fuel acceleration vector acting on the tanker.

The relative equations of motion are derived in the RSW reference frame of the target satellite, whose origin is at located the target satellite's center of mass, its x-axis ($\hat{\boldsymbol{x}}$) is aligned with the target satellite's position vector ($\boldsymbol{r}$), the z-axis towards the orbital angular momentum vector of the target satellite ($\hat{\boldsymbol{w}}$), and the y-axis ($\hat{\boldsymbol{y}}$) forms the right-handed coordinate system [70]. It should be noted that all vectors throughout the paper are considered column matrices and written in bold and italic (e.g., $\boldsymbol{x}$). Moreover, the magnitude of a vector is represented by just italic characters (e.g., x).

The attitude dynamics and kinematics of the tanker spacecraft can be modeled as follows [71].

$$\begin{aligned} \mathbf{I}\dot{\boldsymbol{\omega}} &= \boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega} + \overbrace{\boldsymbol{\tau_c} + \boldsymbol{\tau_s}}^{\tau} \\ \dot{\mathbf{q}} &= \frac{1}{2}\begin{bmatrix} q_4\boldsymbol{\omega} - \boldsymbol{\omega} \times \boldsymbol{q} \\ -\boldsymbol{\omega}^T\boldsymbol{q} \end{bmatrix} \end{aligned} \tag{2}$$

in which $\boldsymbol{\omega}$ is the angular velocity vector, $\mathbf{I}$ represent the moment of inertia of the body, $\mathbf{q} = [\boldsymbol{q} \quad q_4]$ is the quaternion vector, $\boldsymbol{\tau_c}$ is the controlling torque generated by thrusters, and $\boldsymbol{\tau_s}$ represents the torque generated by the fuel motion. In this work, the fuel disturbances terms ($\boldsymbol{d_s}$ and $\boldsymbol{\tau_s}$) are obtained from CFD simulations using OpenFOAM, which solves the Navier–Stokes equations with a Volume-of-Fluid (VOF) formulation for two-phase flow [72]. This approach captures nonlinear and mode-dependent sloshing dynamics more accurately than simplified analytical models.

Consider the undisturbed tanker's nonlinear equations of motion as

$$\dot{\boldsymbol{s}}(t) = \boldsymbol{f}(\boldsymbol{s}(t), \boldsymbol{u}(t)) \tag{3}$$

where $\boldsymbol{s} = [\delta\boldsymbol{r} \quad \delta\dot{\boldsymbol{r}} \quad \mathbf{q} \quad \boldsymbol{\omega}]^{\mathrm{T}}$ denotes the system state vector, $\boldsymbol{u} = \boldsymbol{F_c}$ represents the control input, and $\boldsymbol{f}(\cdot,\cdot) : (\mathbb{R}^n, \mathbb{R}^m) \longrightarrow \mathbb{R}^n$ denotes a nonlinear function of the system. n and m are the number of system states and control inputs, respectively.

Let $\boldsymbol{s}'(t)$ denote the system state in a finite upcoming time horizon, obtained by applying $\boldsymbol{u}(t) = \boldsymbol{u}(t^-)$ to (3) having the initial condition $\boldsymbol{s}'(t) = \boldsymbol{s}(t)$, where $\boldsymbol{u}(t^-)$ corresponds to the most recently applied control command to the system prior to time t. Using the first order approximation of the system (3) around $\boldsymbol{s}'(t)$ and $\boldsymbol{u}(t^-)$, the resulting Linear Time-Varying (LTV) system is as

$$\dot{\hat{\boldsymbol{s}}}(t) = \mathbf{A}_t\hat{\boldsymbol{s}}(t) + \mathbf{B}_t\boldsymbol{u}(t) + \boldsymbol{d}(t) \tag{4}$$

where $\mathbf{A}_t \in \mathbb{R}^{n\times n}$ and $\mathbf{B}_t \in \mathbb{R}^{n\times m}$ are defined as follows

$$\mathbf{A}_t = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{s}}\right|_{\boldsymbol{s}'(t),\boldsymbol{u}(t^-)}, \mathbf{B}_t = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_{\boldsymbol{s}'(t),\boldsymbol{u}(t^-)} \tag{5}$$

and $\boldsymbol{d}(t) \in \mathbb{R}^{n\times 1}$ is defined as

$$\boldsymbol{d}(t) = \dot{\boldsymbol{s}}'(t) - \mathbf{A}_t\boldsymbol{s}'(t) - \mathbf{B}_t\boldsymbol{u}(t^-). \tag{6}$$

The LTV system (4) can be utilized over a finite time horizon to approximate the nonlinear system (3) in case the effect of the fuel sloshing is neglected. This model is used to generate an optimum trajectory in the next chapter using MPC method.

3. MPC framework

As previously noted, the control strategy must ensure satellite safety by incorporating collision avoidance constraints while addressing disturbances such as sloshing, which can cause undesirable velocities during docking. However, modeling the sloshing dynamics directly in the control framework is challenging due to its complexity and the reliance on CFD for accurate simulation. To address this, the control scheme is designed to generate an optimal nominal trajectory for the docking phase based on unperturbed dynamics, while the management of sloshing disturbances is handled separately through the RL component of the framework, discussed later.

MPC is particularly well-suited for this purpose due to its ability to predict system behavior, handle constraints, and optimize performance in real time. MPC has shown a great performance in different aerospace applications including satellite attitude control [73], satellite collision avoidance maneuver design [74], satellite station keeping [75], asteroid hovering [76], quadrotor trajectory control [77], and space tether control [78,79]. By leveraging a mathematical model of the system, MPC predicts future behavior and determines control actions that minimize a predefined cost function while respecting system constraints. This capability enables the generation of collision-free reference trajectories critical for safe docking.

To implement MPC, a model of the system is needed to accurately represent its dynamics and constraints. Using this model, the MPC algorithm predicts the system's future behavior based on the current state and input conditions. These predictions are then used to optimize performance criterion, which could involve minimizing tracking errors or control effort. The optimized solution informs the control actions applied to the system, ensuring it adheres to the desired trajectory. After executing the control action, the system's state is measured and incorporated into the algorithm to refine subsequent predictions. This process of prediction, optimization, and control is repeated at regular intervals, enabling real-time responsiveness to evolving conditions.

In this study, MPC is utilized to formulate a path-planning problem for the docking phase, ensuring the optimum trajectory is free of collisions. The optimization problem solved at each time step can be expressed as follows:

$$\begin{aligned} &\min_{\boldsymbol{u}(t)} \int_{t=t_0}^{t_f} \left(\left\|\widehat{\boldsymbol{s}}(t)-\boldsymbol{s}_f\right\|_{\boldsymbol{\Omega}}^2+\|\boldsymbol{u}(t)\|_{\mathbf{R}}^2\right)dt \\ &s.t. \quad \dot{\widehat{\boldsymbol{s}}}(t)=\mathbf{A}_t\widehat{\boldsymbol{s}}(t)+\mathbf{B}_t\boldsymbol{u}(t)+\boldsymbol{d}(t) \\ &\qquad\qquad \widehat{\boldsymbol{s}}(t)\in\boldsymbol{\Xi} \\ &\qquad \boldsymbol{u}(t)\in\boldsymbol{\upsilon} \quad \widehat{\boldsymbol{s}}(t_0)=\boldsymbol{s}(t_0) \end{aligned} \tag{7}$$

where t_f denotes the final docking time, t_0 denotes the current time instant, $\boldsymbol{s}_f$ represents the desired final docking states, respectively, and $\| .\|_{\boldsymbol{\Omega}}^2$ denotes the weighted norm of "." defined by $(.)^{\mathrm{T}}\boldsymbol{\Omega}(.)$ with $\boldsymbol{\Omega}$ being a positive definite matrix. The system is also subjected to state constraint, $\boldsymbol{\Xi}\in\mathbb{R}^n$, and control input constraint, $\boldsymbol{\upsilon}\in\mathbb{R}^m$, that are convex polytope sets. Given that the optimization problem (7) is a convex quadratic programming problem, it can be efficiently solved using standard Quadratic Programming (QP) solvers [80].

MPC generates an optimal trajectory for the docking phase by minimizing a cost function that balances trajectory tracking accuracy and control effort. This trajectory provides the nominal reference for the docking process. To manage disturbances caused by sloshing, the RL module complements this control strategy, ensuring robust and adaptive performance even under uncertain conditions. Together, MPC and RL form an integrated framework capable of addressing the complex requirements of satellite docking.

4. RL methodology

The RL framework models the agent-environment interaction as an Markov Decision Process (MDP), formally defined as a tuple $M = (S,A,P,R,\gamma)$, where S is the state space, representing all possible system configurations, A is the action space, encompassing all feasible control inputs, $P\,(s_{t+1}|s_t,a_t)$ is the state transition probability, governing the likelihood of transitioning from state s_t to s_{t+1} under action a_t, $R(s_t,a_t)$ is the reward function, defining the immediate reward for executing action a_t in state s_t, and $\gamma\in(0,1]$ is the discount factor, determining the importance of future rewards relative to immediate ones [81]. The agent interacts with the environment by selecting actions according to a policy $\pi(a_t|s_t)$, which dictates the probability of choosing action a_t given state s_t.

4.1. Proximal Policy Optimization (PPO)

PPO leverages an actor-critic framework, where the actor represents the policy, and the critic evaluates actions through a value function. The state-value function $V_{\boldsymbol{w}}^{\pi}(\boldsymbol{s}_t)$ is expressed as [82]:

$$V_{\boldsymbol{w}}^{\pi}(\boldsymbol{s}_t)=\mathrm{E}_{\pi}\left(\sum_{k=t}^{T}\gamma^{k-t}r_k(\boldsymbol{s}_k,\boldsymbol{a}_k)\middle|\boldsymbol{s}_t\right) \tag{8}$$

where $\boldsymbol{w}$ parameterizes the value function. The advantage function $A_{\boldsymbol{w}}^{\pi}(\boldsymbol{s}_t,\boldsymbol{a}_t)$ quantifies the quality of actions relative to the expected value:

$$A_{\boldsymbol{w}}^{\pi}(\boldsymbol{s}_t,\boldsymbol{a}_t)=\left(\sum_{k=t}^{T}\gamma^{k-t}r_k(\boldsymbol{s}_k,\boldsymbol{a}_k)\right)-V_{\boldsymbol{w}}^{\pi}(\boldsymbol{s}_t) \tag{9}$$

The PPO loss function incorporates a policy probability ratio $p_t(\boldsymbol{\alpha})$:

$$p_t(\boldsymbol{\alpha})=\frac{\pi_{\boldsymbol{\alpha}}(\boldsymbol{a}_t|\boldsymbol{s}_t)}{\widehat{\pi}_{\boldsymbol{\alpha}}(\boldsymbol{a}_t|\boldsymbol{s}_t)} \tag{10}$$

This ratio compares the probability of selecting an action before and after a policy update. The PPO objective function is defined as

$$\mathscr{L}(\boldsymbol{\alpha})=\mathrm{E}_{p(\tau)}\left[\min\begin{pmatrix} p_t(\boldsymbol{\alpha})A_{\boldsymbol{w}}^{\pi}(\boldsymbol{s}_t,\boldsymbol{a}_t), \\ \mathrm{clip}[p_t(\boldsymbol{\alpha}),\epsilon]A_{\boldsymbol{w}}^{\pi}(\boldsymbol{s}_t,\boldsymbol{a}_t)\end{pmatrix}\right] \tag{11}$$

where the clipping function constrains updates to ensure stability. It is given by

$$\mathrm{clip}[p_t(\boldsymbol{\alpha}),\epsilon]=\begin{cases} 1-\epsilon\; p_t(\boldsymbol{\alpha})<1-\epsilon \\ 1+\epsilon\; p_t(\boldsymbol{\alpha})<1+\epsilon \\ p_t(\boldsymbol{\alpha}) \quad \text{otherwise}\end{cases} \tag{12}$$

imposes bounds on the policy probability ratio using a clipping parameter ϵ within $(0,1)$.

To improve the state-value function $V_{\boldsymbol{w}}^{\pi}(\boldsymbol{x}_t)$, PPO minimizes the mean squared error loss:

$$L(\boldsymbol{w})=\frac{1}{2}\mathrm{E}_{p(\tau)}\left[\left(V_{\boldsymbol{w}}^{\pi}(\boldsymbol{s}_t)-\left[\sum_{k=t}^{T}\gamma^{k-t}r(\boldsymbol{s}_t,\boldsymbol{a}_t)\right]\right)^2\right] \tag{13}$$

The gradients for updating the policy and value function are computed using the following update rules:

$$\begin{aligned} \boldsymbol{\alpha}_+ &= \boldsymbol{\alpha}_- + \beta_{\alpha}\nabla_{\alpha}J(\boldsymbol{\alpha})|_{\alpha=\alpha_-} \\ \boldsymbol{w}_+ &= \boldsymbol{w}_- - \beta_{\boldsymbol{w}}\nabla_{\boldsymbol{w}}L(\boldsymbol{w})|_{\boldsymbol{w}=\boldsymbol{w}_-} \end{aligned} \tag{14}$$

where β_{α} and $\beta_{\boldsymbol{w}}$ represent the learning rates for the policy and value function updates, respectively. These updates facilitate convergence toward an optimal policy and an accurate approximation of the value function, ensuring the effectiveness of the reinforcement learning process.

PPO achieves a balance between exploration and exploitation through its structured optimization approach, mitigating risks of divergence while maintaining simplicity compared to predecessors such as Trust Region Policy Optimization (TRPO) [83]. The algorithm employs a clipping mechanism to restrict the magnitude of policy updates, enhancing stability and reliability during training.

In this implementation of PPO, a strategy inspired by Gaudet et al. [84] is adopted to dynamically adjust learning parameters in response to the observed Kullback-Leibler (KL) divergence between successive policy updates [85]. By dynamically modulating these parameters, this approach maintains the KL divergence close to a desired target value L_d. This ensures that policy updates proceed incrementally and avoids large deviations that could destabilize the learning process. Throughout training, both the clipping threshold ϵ and the learning rate β_{α} are continuously adapted to achieve the desired stability and convergence.

The policy and state-value functions are implemented using feedforward neural networks, parameterized by α and w, respectively. These parameters are updated according to the gradient update rules (14), employing the Adam optimizer [86] for efficient and adaptive learning. The policy is represented as a multivariate Gaussian distribution with a diagonal covariance matrix, facilitating continuous action selection. During training, neural network outputs are scaled using the running mean and standard deviation of state data encountered during learning to ensure stability and consistency. The outputs of the policy network are further scaled such that values of ± 1 correspond to the maximum and minimum allowable thrust or torque.

The actor network for PPO is designed as a feedforward neural network with two hidden layers, consisting of 128 and 64 neurons, respectively. These hidden layers use the hyperbolic tangent (tanh) activation function, ensuring smooth and bounded outputs. The output layer of the actor network consists of either 3 neurons for the 3-DoF laboratory experiments or 6 neurons for the 6-DoF numerical simulations. The output layer employs linear activation to directly map the outputs to the control signals (e.g., thrust and torque).

The critic network, responsible for estimating the state-value function, features a more complex architecture with three hidden layers consisting of 128, 64, and 8 neurons. Similar to the actor network, the hidden layers use the tanh activation function. The output layer of the critic network is a single neuron with linear activation, providing scalar estimates of the state value.

4.2. SAC network architecture

Unlike PPO, which typically employs two neural networks (actor and critic), SAC utilizes five neural networks.

- Policy Network (Actor): Predicts the probability distribution of actions.
- Two Critic Networks: Estimate the action-value function $\boldsymbol{Q}_\phi(\boldsymbol{s}_t, \boldsymbol{a}_t)$ to mitigate overestimation bias.
- Two Target Critic Networks: Stabilize training by providing reliable target values.

This architecture introduces redundancy and stability, with the target networks helping to correct overestimation issues inherent in value-based methods. SAC is widely regarded for its.

- Stability: The use of dual critics and target networks mitigates overestimation and ensures reliable value estimates.
- Sample Efficiency: Entropy regularization promotes exploration, reducing the number of samples required to learn an effective policy.
- Robustness: Its ability to handle continuous control tasks with high-dimensional action spaces makes it suitable for complex engineering applications.

SAC replaces the state-value function with the action-value function, $\boldsymbol{Q}_\phi(\boldsymbol{s}_t, \boldsymbol{a}_t)$, which evaluates the expected return for a specific action. The objective function for the policy network is given by Ref. [87]:

$$J(\theta) = \mathrm{E}_{\boldsymbol{s}_t \sim \boldsymbol{D}}\left[\min_{i=1,2} \boldsymbol{Q}_{\phi_i}(\boldsymbol{s}_t, \widetilde{\boldsymbol{a}}_t)\chi \log \pi_\theta\left(\widetilde{\boldsymbol{a}}_t | \boldsymbol{s}_t\right)\right] \tag{15}$$

where $\boldsymbol{D}$ is the replay buffer storing past experiences, $\widetilde{\boldsymbol{a}}_t$ is a sampled action from the policy π_θ, and χ regulates the influence of the entropy term. This objective combines reward maximization with entropy regularization, ensuring a balance between exploration and exploitation.

The critic networks in SAC minimize the Temporal Difference (TD) error using the following loss function:

$$L(\phi_i) = \mathrm{E}_{\boldsymbol{s}_t \sim \boldsymbol{D}}\left[\left(\boldsymbol{Q}_{\phi_i}(\boldsymbol{s}_t, \widetilde{\boldsymbol{a}}_t) - \boldsymbol{y}_t\right)^2\right] \tag{16}$$

where,

$$y_t = \boldsymbol{r}_t + \gamma\left(\min_{i=1,2} \boldsymbol{Q}'_{\phi_i}(\boldsymbol{s}_{t+1}, \widetilde{\boldsymbol{a}}_{t+1})\chi \log \pi_\theta\left(\widetilde{\boldsymbol{a}}_{t+1} | \boldsymbol{s}_{t+1}\right)\right) \tag{17}$$

Here y_t is the target value for the critic networks, r_t is the immediate reward, Q'_{ϕ_i} are the target critic networks, and $\widetilde{\boldsymbol{a}}_{t+1}$ is an action sampled from the policy at the next state. This formulation ensures stability in training by minimizing errors between predicted and actual returns. The target critic networks are updated using a soft update mechanism:

$$\phi'_i = \omega\phi'_i + (1-\omega)\phi_i \quad i = 1, 2 \tag{18}$$

where $\omega \in [0, 1]$ is the weighting factor. This approach ensures a gradual update of the target networks, reducing oscillations and stabilizing learning.

The policy network consists of three hidden layers with 256, 128, and 64 neurons, using the tanh activation function, and a linear output layer for the action means. A Softplus transformation ensures the standard deviations remain positive. This architecture enables smooth and robust control for continuous action spaces.

Two independent critic networks, each with three hidden layers of 256, 128, and 64 neurons, are used to mitigate overestimation bias by taking the minimum Q-value during training. Both networks use tanh activation in the hidden layers and a linear output layer for scalar Q-values. To stabilize training, SAC employs two target critic networks, which are soft-updated versions of the critics. These target networks provide smoothed Q-value estimates to prevent training oscillations. An adaptive entropy coefficient is also used to balance exploration and exploitation dynamically, which is critical for handling uncertainties like fuel sloshing during 6-DoF docking.

4.3. Reward function

The effectiveness of RL heavily relies on the design of a well-defined reward function, as the learning process aims to maximize this function. In the context of 6-DoF maneuvers, the reward function is designed to address several objectives, including the minimization of state tracking errors, control effort, and disturbances from fuel sloshing, as well as the reinforcement of successful stabilization. Each of these objectives is assigned appropriate weights through carefully selected design coefficients.

This study explores two configurations for the RL framework: 1) a standalone RL method using PPO and SAC, and 2) an integrated RL-MPC approach. Each configuration has its own distinct reward structure tailored to meet the specific learning and control requirements.

In the standalone RL configuration, the RL algorithm is tasked with independently learning an optimal trajectory and control policy to ensure successful docking of the tanker. The reward function is defined as follows:

$$r(\boldsymbol{s}_t, \boldsymbol{a}_t) = -\|\boldsymbol{u}_t\|^2_{\mathrm{P}} + \frac{\Psi_1}{1 + e^{k(t)\delta}} + \frac{\Psi_2}{1 + e^{k(t)\sigma}} - \|\boldsymbol{f}_s\|^2_\Phi - \|\boldsymbol{\tau}_s\|^2_\eta \tag{19}$$

where the first term, $-\|\boldsymbol{u}_t\|^2_{\mathrm{P}}$, penalizes control effort by introducing a quadratic cost on the control inputs, encouraging energy-efficient maneuvers. The second and third terms represent terminal stabilization bonuses. These terms ensure that as the system approaches its stabilization state, the reward increases exponentially, reinforcing successful docking and stabilization. Here, $k(t) > 0$ is a monotonically increasing function that gradually narrows the reward range over time, and σ and δ represent position and rotation angle stabilization errors, respectively. The parameters $\Psi_1, \Psi_2 > 0$ control the magnitude of the terminal bonuses. The last two terms, $-\|\boldsymbol{f}_s\|^2_\Phi$ and $-\|\boldsymbol{\tau}_s\|^2_\eta$, penalize forces and torques induced by fuel sloshing. By limiting these quantities, the reward function encourages a stable state for the liquid fuel during the maneuver, thereby minimizing unwanted movements caused by sloshing. This design ensures that the RL agent learns to balance the competing objectives of minimizing control effort, achieving stabilization, and maintaining fuel stability.

The integrated RL-MPC configuration introduces the output of MPC into the reward function to accelerate the RL training process. By incorporating an optimal trajectory from MPC, the RL agent is guided toward effective stabilizing trajectories, significantly reducing the learning time. The reward function in this framework is expressed as:

$$r(\mathbf{s}_t, \mathbf{a}_t) = -\|\dot{\hat{\mathbf{s}}}_t - \dot{\mathbf{s}}_t\|_{\mathrm{M}}^2 - \|\mathbf{u}_t\|_{\mathrm{P}}^2 + \frac{\Psi_1}{1 + e^{k(t)\delta}} + \frac{\Psi_2}{1 + e^{k(t)\sigma}} - \|\mathbf{f}_s\|_{\Phi}^2 - \|\mathbf{\tau}_s\|_{\eta}^2 \tag{20}$$

where the first term, $-\|\dot{\hat{\mathbf{s}}}_t - \dot{\mathbf{s}}_t\|_{\mathrm{M}}^2$, quantifies the quadratic weighted error between the state derivatives generated by the RL agent ($\dot{\mathbf{s}}_t$) and the reference trajectory provided by MPC ($\dot{\hat{\mathbf{s}}}_t$). This term serves as a key component in enabling the RL agent to learn effectively from the optimal MPC outputs. The remaining terms are identical to those in the standalone RL configuration, addressing control effort, stabilization bonuses, and sloshing mitigation. By leveraging the MPC trajectory as a reference, the RL agent receives a clear reward signal across the entire state-space, allowing it to efficiently learn a control strategy that not only achieves stabilization but also adheres to optimality.

It is important to note that while MPC trajectories are used in the RL reward function to accelerate convergence, the RL agent does not inherit the limitations of the simplified MPC dynamics in Eq. (7). Instead, RL continually adapts its policy based on feedback from CFD-based simulations of sloshing disturbances. This ensures that the final control solution is not restricted to the nominal MPC model, but rather integrates both the computational efficiency of MPC and the robustness of data-driven learning.

The standalone RL configuration offers the advantage of independent learning, where the RL algorithm autonomously determines optimal trajectories and control actions. However, this approach may require extensive training time to converge to a reliable policy. The integrated RL-MPC configuration, shown in Fig. 4, mitigates this limitation by accelerating the training process through the infusion of MPC-generated trajectories, thus combining the adaptability of RL with the precision and robustness of MPC. Both reward functions emphasize key objectives such as energy efficiency, stabilization accuracy, and fuel sloshing mitigation, making them well-suited for the complex task of satellite docking.

5. Experimental validation

This section presents the experimental validation of the proposed methodologies, designed to evaluate the effectiveness of integrating MPC with RL. Two distinct scenarios are analyzed to provide comprehensive insights into the system's performance.

In the first scenario, experiments are conducted in the Zero-G Lab to validate the benefits of integrating RL with MPC. The lab environment, featuring a planar floating platform, provides a controlled setup to study stabilization performance under disturbances. While this scenario does not incorporate fuel sloshing effects due to the practical limitations of recreating sloshing dynamics in a laboratory environment, it serves as a proof of concept to demonstrate that the integration of MPC can enhance RL performance in terms of convergence speed, robustness, and stabilization accuracy. The objective of the first scenario is to design a stabilizer for the floating platform.

The second scenario simulates an actual satellite docking operation in orbit, incorporating the complexities of 6-DoF dynamics and fuel sloshing disturbances. As it is impractical to recreate such conditions in the lab, this scenario is investigated through a high-fidelity numerical environment. Four control configurations—PPO, SAC, PPO-MPC, and SAC-MPC—are analyzed to evaluate the robustness and adaptability of the proposed methods under real-world conditions.

The combination of physical experimentation and numerical simulation ensures a comprehensive evaluation of the proposed approaches. The lab experiments validate the core concept of MPC-RL integration in a simplified setup, while the numerical simulations extend this validation to a more complex and realistic docking scenario.

It should be noted that the Zero-G Lab experiments were not intended to replicate propellant sloshing phenomena. The floating platform and air-bearing system provide a planar microgravity emulation environment for validating the spacecraft's pose control. All propellant sloshing effects in this study were instead modeled using CFD-based simulations under microgravity conditions. These simulations generated nonlinear disturbance forces and moments that were subsequently incorporated into the RL training process. Therefore, the experimental campaign and the CFD simulations serve complementary purposes: the former demonstrates the feasibility of the control law in hardware, while the latter ensures robustness against realistic fluid–structure interactions.

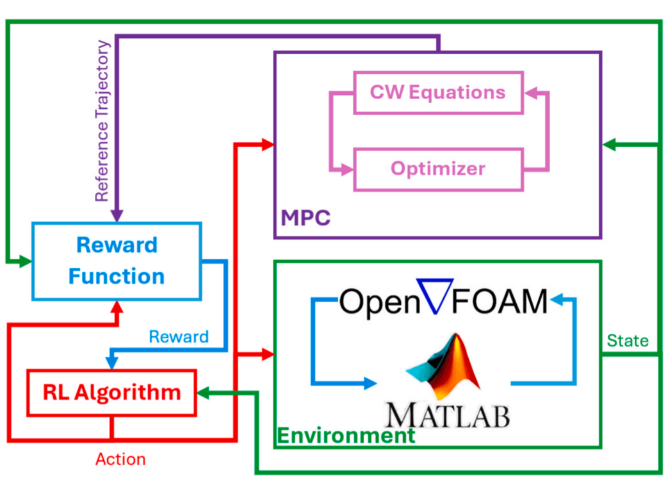

Fig. 4. The schematic diagram of the integrated RL-MPC method.

5.1. *Case 1: planar stabilization in the Zero-G lab*

The initial experiments were carried out in the Zero-G Lab, a specialized facility designed to simulate near-frictionless motion. The primary objective was to develop a control strategy to stabilize the floating platform at the origin, located at the center of the lab's floor. The lab features a large testing area measuring 5 m × 3 m × 2.3 m, equipped with floating platforms that move effortlessly over an epoxy-coated floor. Near-frictionless motion is achieved through the use of air-bearings, which expel high-pressure air onto the floor surface to eliminate mechanical contact. This setup provides an ideal environment for investigating planar stabilization dynamics. The platforms are actuated by eight nozzles, capable of generating forces along two translational axes (X and Y) and one rotational axis (Z). Each nozzle can produce up to 1 N of force when operated at a pressure of 10 bars.

The motion of the floating platforms is monitored using an OptiTrack motion capture system, which consists of six Prime 13W cameras operating at a frequency of 240 Hz. An active marker is mounted at the center of the platform's top plate to facilitate precise tracking of its position and orientation. These platforms are seamlessly integrated into the Robot Operating System (ROS) network, and a ROS-MATLAB bridge allows programming and control using MATLAB, enabling efficient experimentation and evaluation. More information about the Zero-G Lab can be found in Refs. [88–91].

5.1.1. *Training setup*

The RL algorithm was implemented in MATLAB and trained over a maximum of 20,000 iterations to ensure network convergence. During the training phase, data was collected in batches of 200 episodes, and updates to the policy and state-value functions. The policy generated force and torque commands at discrete time intervals of 0.1 s. Training episodes were limited to a duration of 60 s to enable efficient data collection and stabilization. For testing, this limit was extended to 100 s to ensure the system had sufficient time to complete valid stabilization trajectories.

The stabilization criteria required strict adherence to predefined thresholds: a position accuracy of within 0.05m, velocity within ± 0.1 m/s, rotational accuracy within $\pm 5^\circ$, and angular velocity not exceeding $\pm 1^\circ$ per second. One of the primary objectives was to develop a robust feedback control policy capable of handling a wide range of initial conditions, thereby ensuring adaptability and robustness against uncertainties. The training parameters used in the study are outlined in Table 1.

5.1.2. Training results

Fig. 5 illustrates the normalized average cumulative reward achieved during training. The results demonstrate that the integrated PPO-MPC approach consistently outperformed the PPO-only method. The PPO-MPC configuration exhibited faster convergence rates and higher rewards by the end of the training phase. This improvement can be attributed to the fact that the PPO algorithm benefited from the optimal solutions generated by MPC, which served as a guiding trajectory for RL. This integration significantly reduced the exploratory burden on the RL agent, allowing it to achieve optimal behavior more efficiently. In contrast, the PPO-only method required extensive exploration, which increased the risk of becoming trapped in suboptimal solutions. While the PPO-only method might eventually converge to the performance level of PPO-MPC, this would likely require a substantially longer training period. Notably, as shown in Fig. 5, the PPO-only configuration had not fully converged even after 20,000 episodes, underscoring the benefits of integrating MPC for faster convergence.

5.1.3. Experimental results and discussion

Following training, both the PPO-MPC and PPO-only methods were tested in real-world experiments involving the floating platform in the Zero-G Lab [65]. During these experiments, the platform was manually disturbed at four distinct time intervals, requiring the control algorithm to autonomously restore stabilization at the center of the lab. The disturbances were introduced to test the adaptability and robustness of the control policies, and their timing is marked with red arrows in the corresponding figures.

Fig. 6 illustrates the progression of the floating platform during a test, with snapshots captured at equal time intervals. Initially, the platform is in a stabilized condition, maintaining its position and orientation. Following this, a manual disturbance is applied, displacing the platform from its initial state. The platform begins to drift away, eventually reaching its maximum deviation from the stabilization point. The control system then activates, reversing the platform's direction and guiding it back toward the stabilization state. As time progresses, the platform gradually reduces its positional and angular errors, moving closer to the desired state. By the final snapshot, the platform successfully returns to full stabilization, maintaining its position and orientation despite the earlier disturbance. This sequence effectively demonstrates the control strategy's ability to restore and sustain stabilization under external perturbations.

The performance of the PPO-MPC method is shown in Fig. 7, in which $dt = 0.1$ second. Each time the platform was disturbed, the PPO-MPC approach successfully restored stabilization, maintaining position

Table 1
The PPO training setting parameters.

Parameter	Value
KL_d	0.001
γ	0.98
M	diag(1, 10, 5)
P	diag(10, 10, 10)
Ψ_1	100
Ψ_2	10

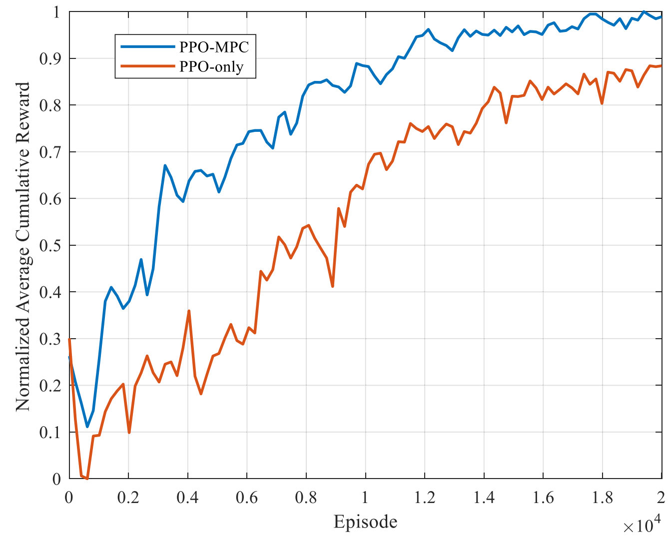

Fig. 5. The normalized average cumulative reward in the training phase of Case 1.

and orientation errors within the predefined thresholds. For instance, while the second disturbance resulted in a larger deviation in rotation angle, the platform quickly returned to its stabilization state. Other disturbances primarily affected position, with minimal impact on orientation. The actuation signals generated by the thrusters, as depicted in Fig. 8, demonstrate the efficient application of pulse-width pulse-frequency (PWPF) modulation.

In contrast, the PPO-only method exhibited slower stabilization and larger errors. As illustrated in Fig. 9, while the platform returned to its origin after each disturbance, the stabilization errors exceeded the defined thresholds. Specifically, the position error averaged around 0.15m, and the rotational error reached approximately 10°. This outcome aligns with the training phase results, where the PPO-only method achieved lower rewards compared to PPO-MPC. The thruster actuation for the PPO-only configuration, shown in Fig. 10.

These experimental results underscore the effectiveness of integrating PPO with MPC. The integrated approach not only achieved precise and rapid stabilization under disturbance but also demonstrated superior convergence properties compared to standalone RL methods. The findings validate the hypothesis that leveraging MPC as a guiding trajectory for RL significantly enhances overall performance.

5.2. Case 2: docking scenario in a real-world environment

In this scenario, a realistic space environment is simulated to evaluate the robustness and adaptability of the proposed control methods. Four configurations—PPO, SAC, PPO-MPC, and SAC-MPC—are analyzed to determine their effectiveness in handling docking maneuvers under real-world conditions. The test involves a LEO) satellite positioned in a circular sun-synchronous orbit at an altitude of 600 km, selected as the target satellite for refueling. The docking tanker is assumed to operate within the same orbital plane, initially positioned within a 200×120 m admissible zone (expressed in the RSW reference frame) around the target satellite prior to the docking phase, as depicted in Fig. 11. In the figure, the blue marker represents the tanker, while the green marker denotes the target satellite.

To ensure safe docking, certain constraints are imposed. The tanker is required to maintain a minimum distance of 10 m from the target satellite to allow for potential rotational adjustments around the target without risking collision. Additionally, the tanker must remain within the admissible zone to preserve the validity of the CW equation (1) used during the docking phase. The tanker is equipped with a partially filled spherical fuel tank of 1 m radius and thrusters capable of producing a

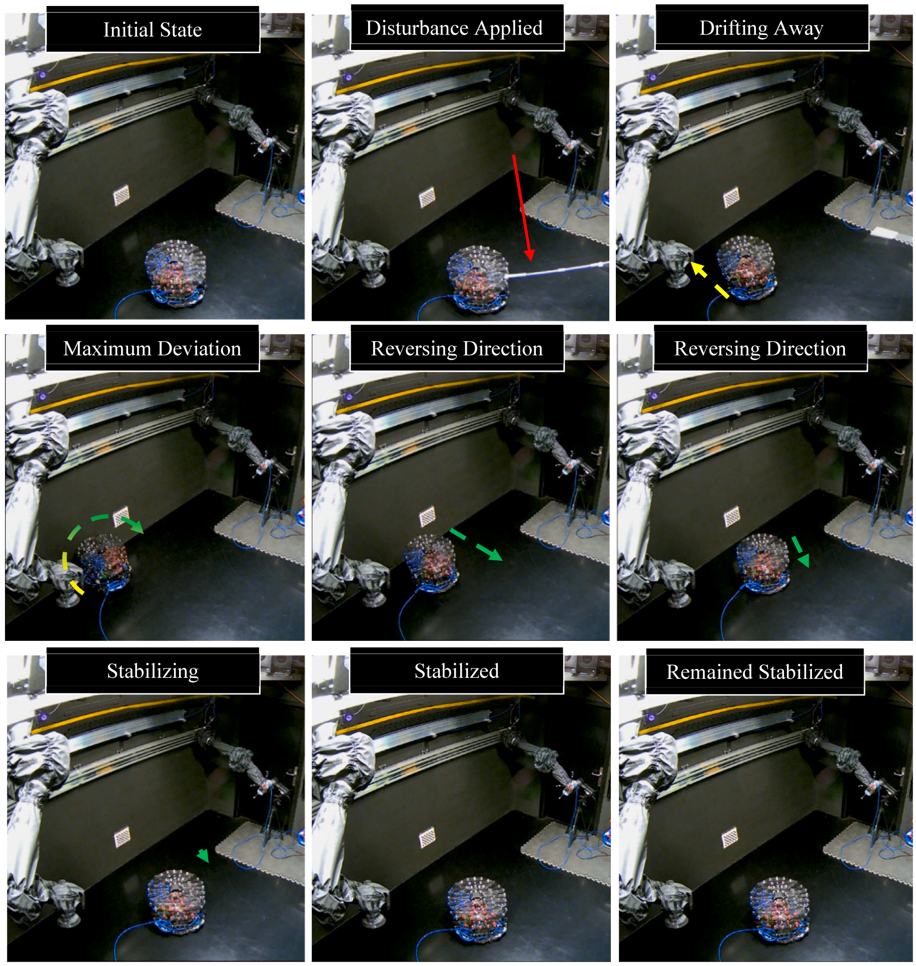

Fig. 6. Time-lapse of the stabilization experiment conducted in the Zero-G Lab.

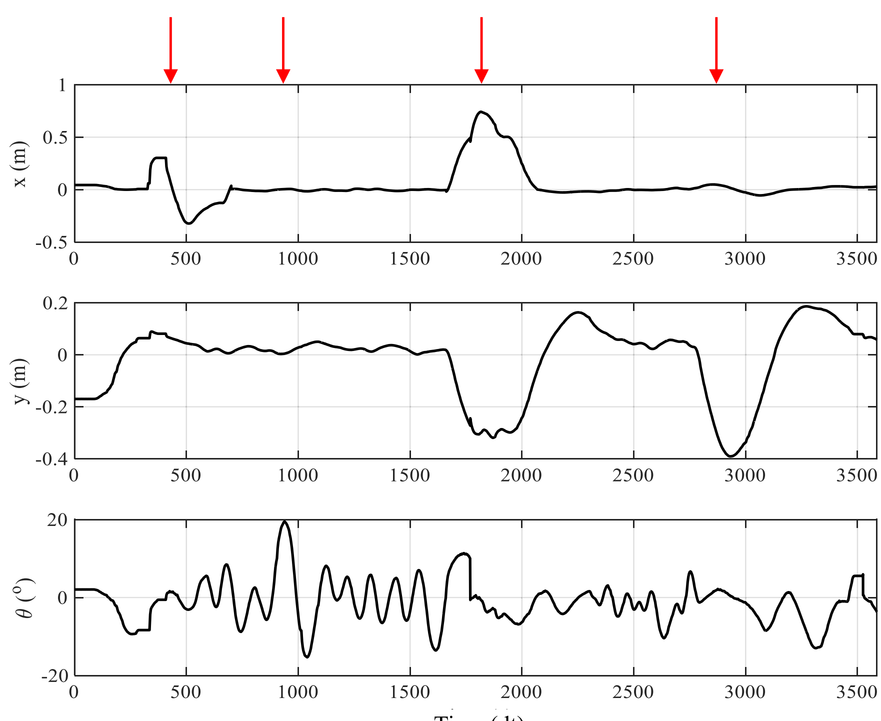

Fig. 7. Performance of the PPO-MPC approach in disturbance rejection at Zero-G Lab experiment.

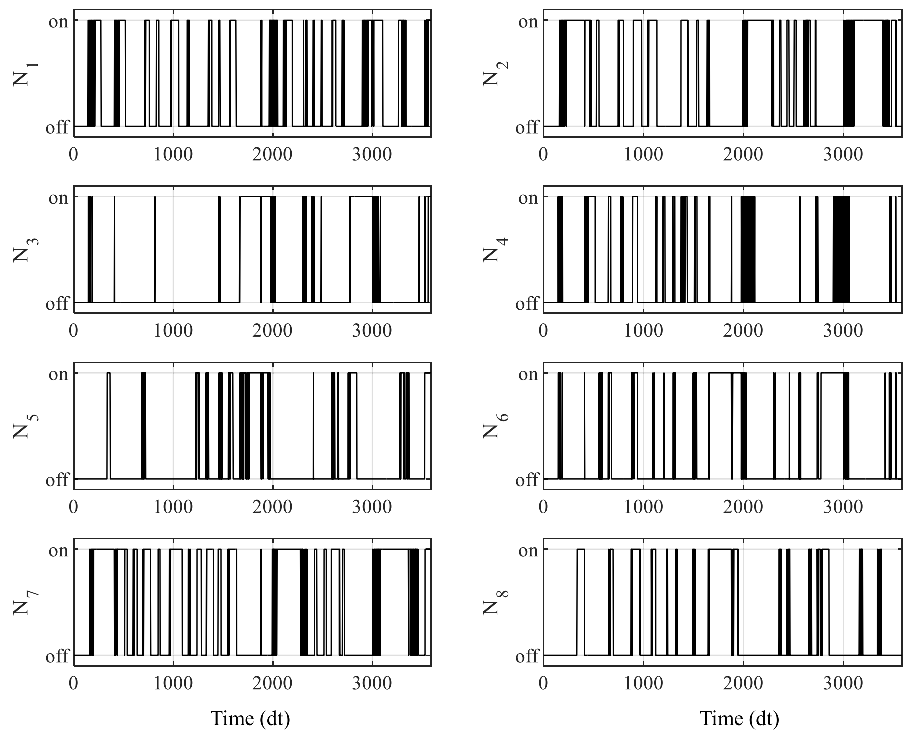

Fig. 8. Thruster actuation profile in PPO-MPC control.

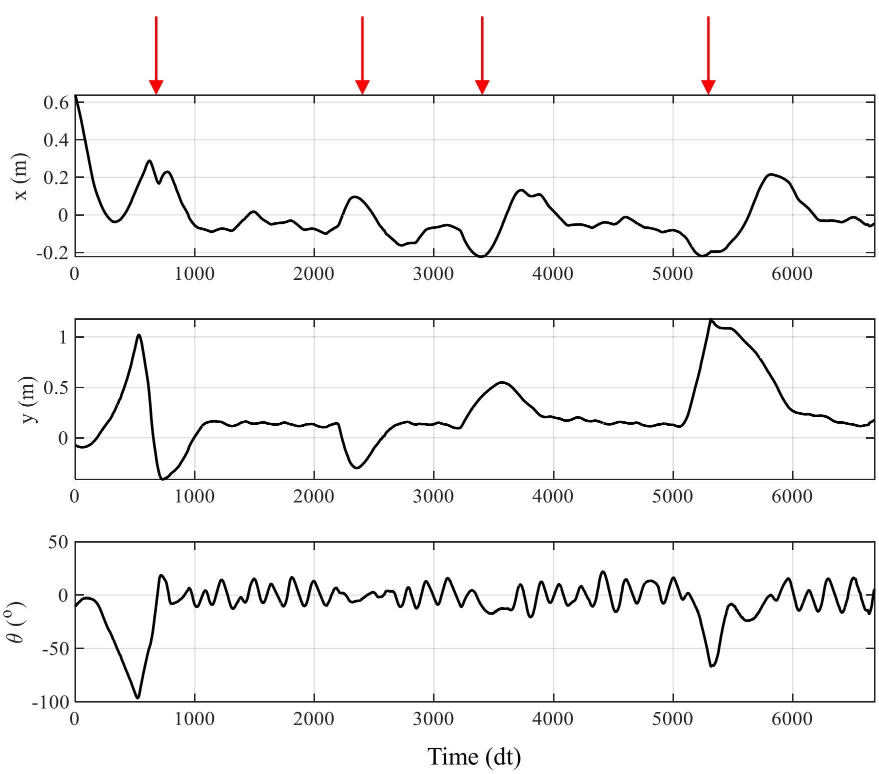

Fig. 9. Performance of the PPO-only approach in disturbance rejection at Zero-G Lab experiment.

maximum force of 10 N. The remaining characteristics of the tanker model are provided in Table 2.

The simulations were conducted under microgravity conditions to replicate the environment experienced during the docking phase. The interFoam solver, which employs the Volume of Fluid method, was selected to accurately model two-phase incompressible flow and track the liquid-gas interface within the tank.

The computational domain was designed as a spherical tank with a radius of 1 m, consistent with the physical dimensions of the tanker. The domain was discretized using a structured mesh comprising approximately 500,000 cells, providing sufficient resolution to capture the liquid-gas interface and interactions near the tank walls. To ensure

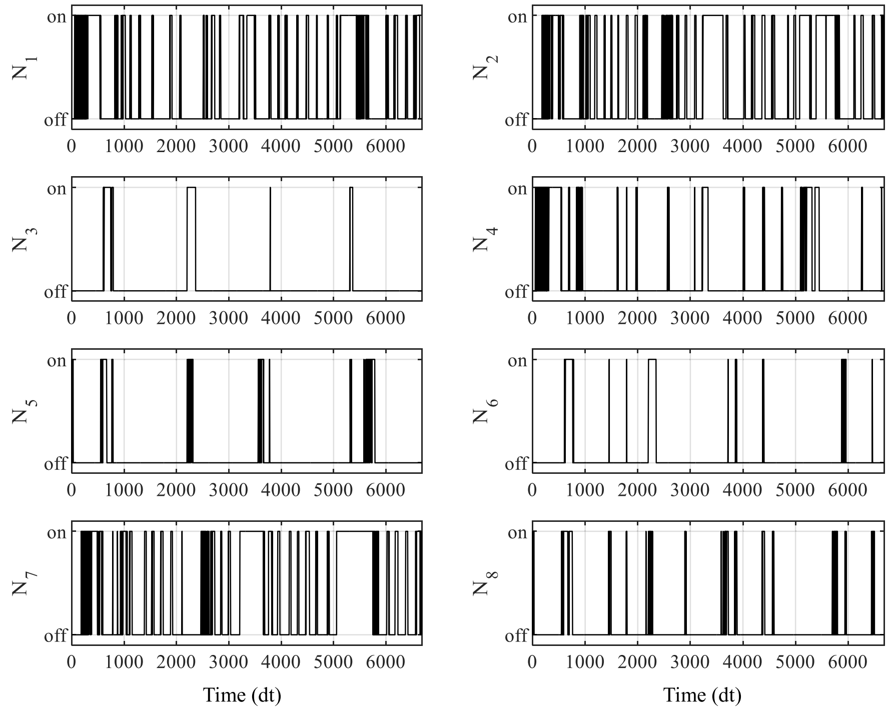

Fig. 10. Thruster actuation profile in PPO-only control.

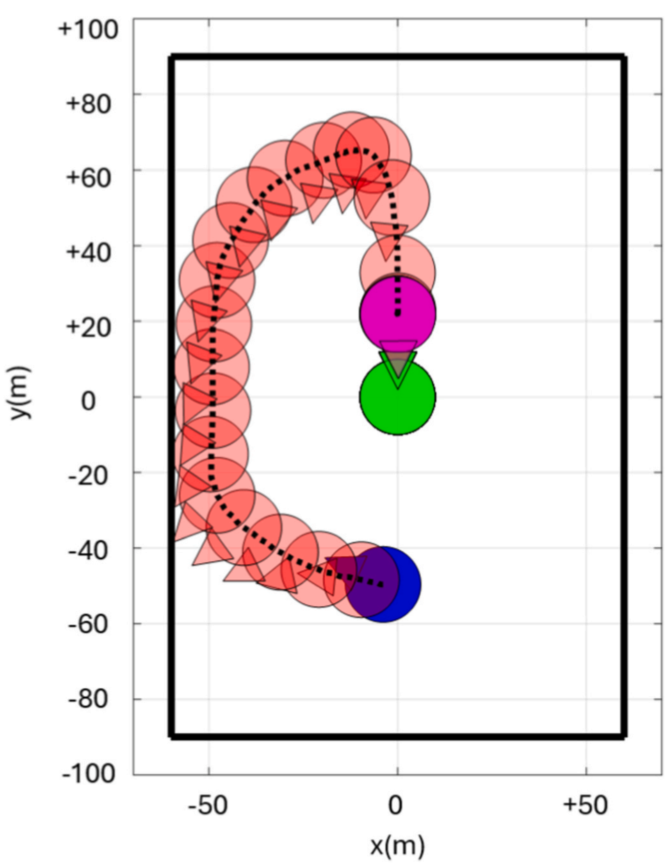

Fig. 11. The docking admissible zone, initial position of the tanker, and the docking trajectory toward the target satellite.

Table 2
The tanker's model physical characteristics.

Parameter	Value	Unit
m	50–300	kg
M	250	kg
l	1	m
b	0.3	m
$\mathbf{I}$	diag(100, 100, 150)	Kg·m^2

realistic boundary behavior, no-slip boundary conditions were applied to the tank walls, while continuity of velocity and pressure was maintained at the liquid-gas interface. An adaptive time-stepping scheme was used to maintain numerical stability, with the Courant number restricted to a maximum value of 0.5. The liquid phase was modeled with properties representative of liquid propellant, specifically a density of 900 $\frac{kg}{m^3}$ and a dynamic viscosity of 0.1 Pa.s, while the gas phase was treated as a vacuum with negligible density and viscosity.

5.2.1. *Training setup*

For training purposes, the initial position and attitude of the tanker are randomly selected within the defined admissible zone. The performance of the proposed methods is also evaluated under varying fuel levels onboard the tanker, ranging from 10 % to 90 % of the tank's capacity, in 20 % intervals. This allows for a comprehensive analysis of how the algorithms perform under different dynamic properties influenced by fuel distribution and sloshing.

The docking control system was implemented through the integration of MATLAB and OpenFOAM. Training was conducted over a maximum of 50,000 iterations to ensure convergence of the RL networks. During training, data was collected in batches of 100 episodes, with updates to the policy and state-value functions. Each training episode was limited to a duration of 5000 s to efficiently gather training data while ensuring proper docking. For testing, the time limit was extended to 10,000 s to provide the system with sufficient time to execute complete and valid docking trajectories.

Strict stabilization criteria were defined for docking success. These criteria required the tanker to achieve a position accuracy of within 0.01 m, a docking velocity of less than 0.05 m/s in the docking direction and less than 0.01 m/s in any other direction, a rotational accuracy within $\pm 0.5^{\circ}$, and an angular velocity not exceeding $\pm 0.01^{\circ}$ per second. The parameters used for training are listed in Table 3.

5.2.2. Training results

Fig. 12 presents the average cumulative reward achieved during the training phase. The results clearly indicate that the integrated RL-MPC approaches (SAC-MPC and PPO-MPC) consistently outperformed their RL-only counterparts (SAC-only and PPO-only). The RL-MPC configurations exhibited faster convergence rates and achieved higher rewards at the end of the training process. This improvement can be attributed to the role of MPC in providing optimal trajectories that guided the RL agent, reducing the exploration burden and enabling it to learn optimal behavior more efficiently. In contrast, the RL-only methods required extensive exploration, increasing the risk of becoming trapped in suboptimal solutions. While the RL-only approaches might eventually converge to the performance level of RL-MPC, doing so would require significantly more training time.

Additionally, the results highlight the superior performance of the SAC-based methods (SAC-MPC and SAC-only) compared to the PPO-based methods (PPO-MPC and PPO-only). Among these, SAC-MPC achieved the highest cumulative reward with the least variability, followed by SAC-only. Similarly, PPO-MPC outperformed PPO-only in both convergence speed and reward magnitude. These findings demonstrate not only the benefits of integrating RL with MPC but also the efficiency of SAC in the training phase, as it exhibited better stability and higher reward consistency compared to PPO-based methods.

Figs. 13 and 14 depict eight sequential frames showing the fuel phase fraction (ρ_w) inside the tank during two different scenarios, captured at constant time intervals. The first set of frames, Fig. 13, corresponds to the case where the control system employs the MPC method without explicitly modeling or limiting the forces and torques generated by the fuel sloshing inside the tank. In contrast, the second set, Fig. 14, illustrates the scenario where these sloshing-induced forces and torques are incorporated into the reward function and actively minimized using the SAC-MPC method.

In the first scenario, Fig. 13, it is evident that fuel sloshing is significant throughout the tanker's movement. This effect becomes particularly pronounced in the final frames, which correspond to the docking phase. During these moments, the fuel inside the tank exhibits substantial motion, inducing forces and torques that could compromise the stability of the tanker during the critical docking maneuver.

In the second scenario, Fig. 14, where the SAC-MPC method incorporates sloshing dynamics into the reward function, a distinct reduction in fuel movement is observed. Although the sloshing is initially considerable, it gradually diminishes over time. By the final frames, corresponding to the docking phase, the fuel is nearly stationary, generating minimal forces and torques on the tanker. This reduced fuel

Table 3
The training setting parameters for case 2.

Parameter	Value
γ	0.98 (PPO)/0.99 (SAC)
KL_d	0.001 (PPO)
β_a	1×10^{-4} (PPO)/1×10^{-4} (SAC)
β_w	1×10^{-4} (PPO)/1×10^{-4} (SAC)
Replay Buffer Size	10^6 (SAC)
Batch Size	200 (PPO)/256 (SAC)
τ	0.005 (SAC)
P	diag(10, 10, 10, 10, 10, 10)
η	10
M	$\mathrm{diag}(1_{1\times3}, 10_{1\times3}, 5_{1\times3}, 50_{1\times3})$
Ψ_1	100
Ψ_2	10
Φ	5

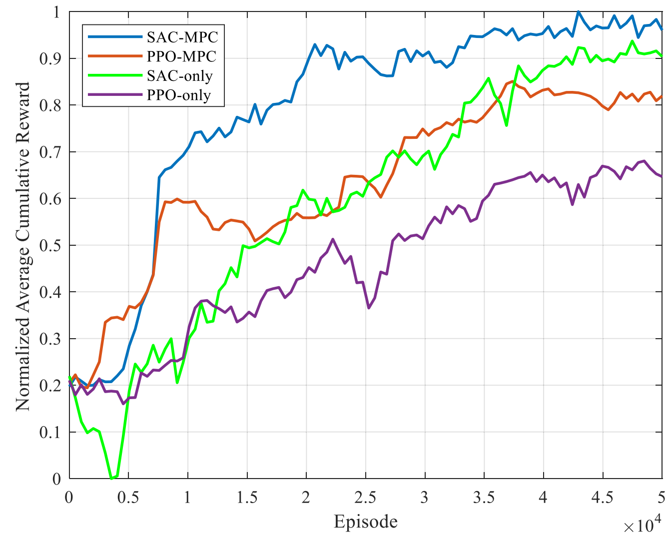

Fig. 12. The normalized average cumulative reward in the training phase of Case 2.

motion ensures a smoother and safer docking process, as it prevents sudden destabilizing movements caused by sloshing forces. The comparison clearly demonstrates the efficacy of the SAC-MPC approach in mitigating sloshing effects and enhancing the overall safety and reliability of the docking operation.

One of the key parameters influencing the performance of the docking system is the amount of fuel inside the tanker's spherical tank. When the tank is completely full (100 %) or empty (0 %), fuel sloshing disturbances are negligible. In the case of an empty tank, no liquid is present to cause sloshing, while a completely full tank prevents the liquid from moving due to the lack of free surface. Under these conditions, the system behaves as a rigid body, and traditional control methods can reliably handle the docking process. However, when the tank is partially filled, the liquid fuel introduces dynamic disturbances through sloshing forces and torques, which vary significantly with the fuel level.

The dynamics of fuel sloshing are highly dependent on the fuel's volume within the tank. At intermediate fuel levels, particularly around 50 % capacity, the liquid has both substantial mass and sufficient free space to move, generating the largest sloshing-induced forces and torques. This creates the most challenging condition for the control system, as the disturbances are not only larger in magnitude but also more dynamic and unpredictable. Fig. 15 compares the success ratios of the four control configurations—PPO, SAC, PPO-MPC, and SAC-MPC—across fuel levels ranging from 10 % to 90 %. The results clearly show a drop in the success ratio for all methods near 50 % capacity, corresponding to the peak disturbances caused by sloshing.

Interestingly, the behavior diverges at higher and lower fuel levels. As the fuel level increases beyond 50 %, the liquid becomes more cohesive, resembling a semi-solid, as its movement becomes restricted by the tank's geometry. This reduces sloshing and results in smaller disturbances, leading to improved performance of the control methods. Conversely, at lower fuel levels below 50 %, while the liquid has more space to slosh, the overall fuel mass is smaller. This results in weaker sloshing forces and torques acting on the tanker body, which also improves the success ratio of the control methods. These trends are consistent with fluid dynamics principles and illustrate the importance of accounting for fuel levels in docking control system design.

The results also highlight the comparative robustness of the control methods. SAC-based methods (SAC-only and SAC-MPC) demonstrate significantly better performance across all fuel levels compared to PPO-based methods (PPO-only and PPO-MPC). Among these, SAC-MPC achieves the highest success ratios, maintaining consistent

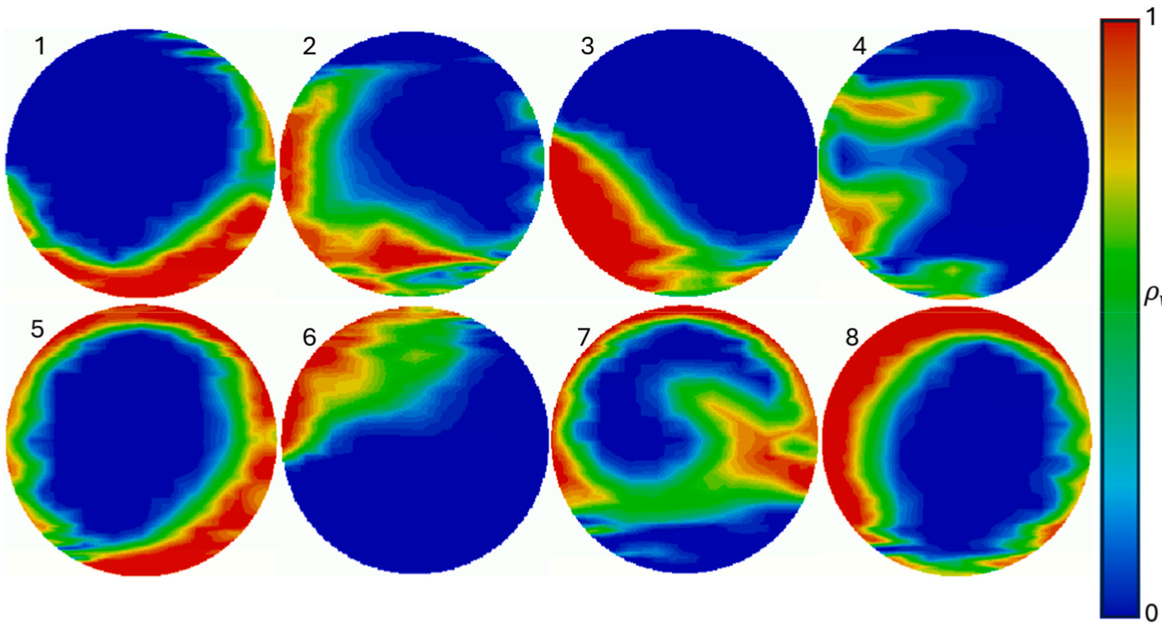

Fig. 13. Fuel phase fraction inside the tank using MPC-only method.

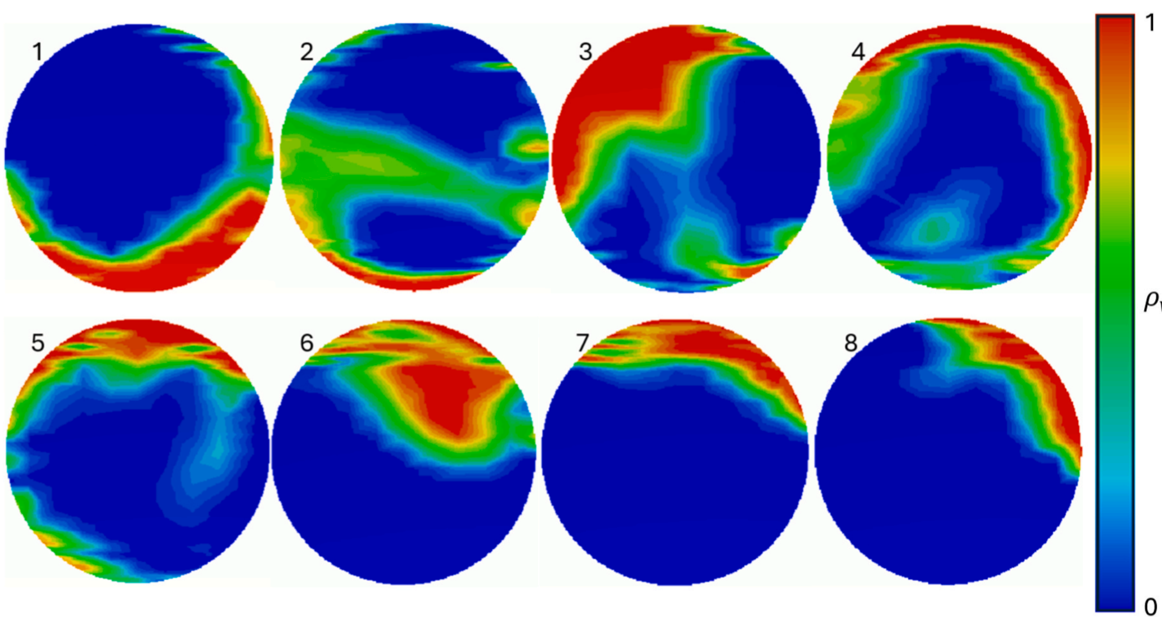

Fig. 14. Fuel phase fraction inside the tank for SAC-MPC method.

performance even in the critical region near 50 % fuel capacity. This robustness can be attributed to SAC's entropy regularization mechanism, which promotes exploratory behavior during training, enabling the policy to handle highly dynamic and uncertain conditions, such as sloshing. In contrast, PPO-based methods exhibit a more substantial performance drop near 50 % capacity, indicating their comparatively lower adaptability to fuel sloshing disturbances.

To evaluate the performance of the control strategies, a comprehensive Monte Carlo simulation was conducted across five configurations: PPO, SAC, PPO–MPC, SAC–MPC, and the classical LQG–MPC method [48]. The simulations considered varying fuel levels (10 %–90 %) and random initial conditions within the admissible zone around the target satellite. Key performance metrics, including success ratio, docking accuracy (position, velocity, and attitude errors), and control effort, were compared across all methods. The inclusion of LQG–MPC, a traditional model–based approach, provides a benchmark to highlight the advantages of the RL–based methods in handling model uncertainties such as fuel sloshing. Table 4 summarizes the average results and variance over 1000 Monte Carlo simulations for each control method.

The success ratio, defined as the percentage of docking attempts that met all predefined thresholds (position, velocity, attitude, and time constraints), highlights the reliability of each method. SAC-MPC achieved the highest success ratio at 96 %, followed by PPO-MPC at 92 %. The standalone RL methods (SAC and PPO) performed worse, with success ratios of 89 % and 78 %, respectively. The smaller variance in SAC-MPC and PPO-MPC indicates greater consistency and robustness compared to the standalone methods. The classical LQG–MPC method showed the lowest success ratio (59 %), confirming its sensitivity to sloshing uncertainties. This demonstrates that the integration of MPC

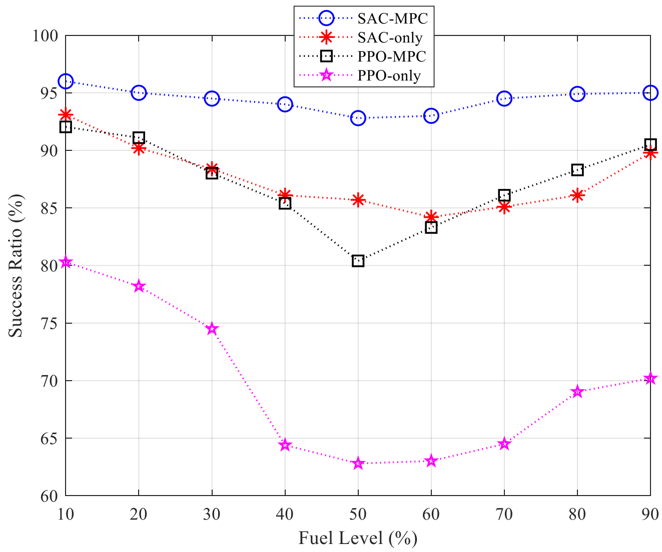

Fig. 15. Success ratio comparison of different control methods across fuel levels.

Table 4
Performance metrics of control methods based on Monte Carlo simulations.

Metric	PPO	SAC	PPO-MPC	SAC-MPC	LQG-MPC [48]
Success Ratio (%)	78	89	92	96	59
Position Error (m)	0.06 ± 0.03	0.05 ± 0.02	0.04 ± 0.01	0.02 ± 0.01	0.1 ± 0.05
Velocity Error (m/s)	0.03 ± 0.01	0.02 ± 0.01	0.02 ± 0.005	0.005 ± 0.003	0.06 ± 0.03
Attitude Error (deg)	0.5 ± 0.2	0.5 ± 0.15	0.4 ± 0.1	0.2 ± 0.08	0.6 ± 0.3
Ang. Vel. Error (deg/s)	0.02 ± 0.01	0.02 ± 0.005	0.02 ± 0.003	0.005 ± 0.002	0.03 ± 0.02
Normalized Control Effort	0.86 ± 0.12	0.80 ± 0.1	0.61 ± 0.08	0.59 ± 0.07	0.91 ± 0.6

significantly improves the reliability of the RL framework, as the MPC component provides optimal guidance, reducing the burden on RL's exploration phase.

Regarding position and velocity errors, SAC-MPC achieved the lowest average position (0.02 ± 0.01 m) and velocity errors (0.005 ± 0.003 m/s), followed by PPO-MPC with slightly higher values (0.04 ± 0.01 m and 0.02 ± 0.005 m/s, respectively). SAC-only and PPO-only methods exhibited larger position and velocity errors with greater variance, indicating less precise control. The smaller variance of SAC-MPC shows that it not only achieves higher accuracy but does so consistently across a wide range of scenarios. While LQG–MPC exhibited the largest errors, indicating its inability to compensate for unmodeled sloshing dynamics.

Regarding attitude and angular velocity errors, SAC-MPC demonstrated superior control, with an average attitude error of $0.2 \pm 0.08^\circ$ and an angular velocity error of 0.005 ± 0.002 deg/s. PPO-based methods showed significantly larger errors, with PPO-only having the highest average attitude error ($0.5 \pm 0.2^\circ$) and angular velocity error (0.02 ± 0.01 deg/s). The reduced variance in SAC-MPC's angular errors highlights its ability to maintain consistent orientation and rotational stability, which is critical for precise docking.

The Normalized Control Effort (NCE) provides a fair comparison of the control strategies by adjusting the control effort in each Monte Carlo trial relative to the maximum control effort required by any method for that specific trial. This means that for each trial, the total thrust by each method is computed, and then the values are normalized against the highest control effort observed across all methods in that same trial. This ensures that variations in initial conditions are accounted for, preventing biases that could arise if different methods were tested under significantly different energy demands. By using this approach, NCE provides a relative efficiency score within each trial, making it possible to compare methods across varying docking conditions.

The results show that SAC-MPC consistently required the lowest normalized effort (0.59 ± 0.07), indicating that in each trial, it expended significantly less thrust than the least efficient method. PPO-MPC followed closely at 0.61 ± 0.08, demonstrating improved efficiency due to the use of MPC for trajectory optimization. Standalone RL methods, however, exhibited significantly higher normalized effort values, with SAC-only at 0.80 ± 0.10 and PPO-only at 0.86 ± 0.12, meaning that these methods frequently required close to the maximum control effort observed in each trial among the RL methods. However, LQG–MPC had the highest average effort (0.91 ± 0.6), reflecting the additional thrust needed to maintain stability under uncertain sloshing conditions.

The reduction in control effort in RL-MPC approaches highlights the ability of MPC to optimize control trajectories, resulting in smoother, more energy-efficient docking maneuvers. Furthermore, SAC-MPC not only minimized the normalized effort but also exhibited lower variance, signifying a more stable and predictable performance across different initial conditions. These findings confirm that integrating RL with MPC leads to substantial improvements in control efficiency, making SAC-MPC the most effective method for satellite docking while minimizing fuel consumption and ensuring precise stabilization.

SAC-based methods outperformed PPO-based methods across all metrics. SAC-MPC demonstrated not only higher success ratios and accuracy but also lower variance, indicating its superior robustness to disturbances such as fuel sloshing. This is likely due to SAC's entropy regularization mechanism, which promotes exploratory behavior during training and allows it to generalize better to dynamic and uncertain conditions. PPO, while simpler, showed less adaptability, particularly under challenging conditions such as intermediate fuel levels (e.g., ∼50 %), where sloshing-induced disturbances are greatest.

The integration of MPC into the RL framework significantly improved performance. Both SAC-MPC and PPO-MPC configurations consistently outperformed their standalone counterparts across all metrics. The MPC component enhances the control system by providing predictive trajectories that guide the RL agent, reducing the need for extensive exploration and mitigating the impact of fuel sloshing. SAC-MPC, in particular, achieved the highest overall performance, with better docking precision, lower control effort, and higher robustness across the tested conditions.

6. Conclusion

This study introduced an integrated Model Predictive Control (MPC) and Reinforcement Learning (RL) framework for autonomous satellite docking, addressing the challenges posed by fuel sloshing in microgravity. The results demonstrated that combining MPC's predictive capabilities with RL's adaptability significantly improved docking precision, robustness, and efficiency. Although the MPC component relies on a simplified dynamic model, the final combined SAC-MPC framework does not converge to an MPC solution. Instead, by incorporating CFD-based feedback and leveraging RL's search capability, the framework produces adaptive policies that account for nonlinear and fragmented sloshing behaviors. Thus, the simplifications inherent in MPC do not compromise the robustness of the integrated approach.

Experimental validation in the Zero-G Lab showed that the PPO-MPC framework achieved a faster convergence rate compared to standalone PPO, leading to reduced stabilization errors. We acknowledge that the ground-based setup does not replicate full two-body docking dynamics or continuous sloshing effects. Instead, its role is to validate the integrated control law under hardware conditions, while CFD simulations provide the necessary fidelity for sloshing dynamics. Together, these ensure both practical feasibility and robustness.

In high-fidelity 6-DoF simulations incorporating realistic sloshing effects, the SAC-MPC framework outperformed all other control strategies, achieving a 96 % docking success rate, compared to 92 % for PPO-MPC, 89 % for SAC-only, and 78 % for PPO-only. Additionally, SAC-MPC reduced position and velocity errors by 50 % and 75 %, respectively, compared to PPO-only. Moreover, the normalized control effort was 30 % lower in SAC-MPC compared to PPO-only, highlighting significant efficiency gains in thrust utilization. The ability of SAC-MPC to adapt to varying fuel levels, particularly at 50 % fuel capacity, where sloshing disturbances are most severe, demonstrates its robustness in handling complex docking scenarios. The results confirm that SAC's entropy regularization enhances policy stability, making it the superior choice for real-time aerospace applications. By integrating RL and MPC, this study provides a scalable and fuel-efficient solution for autonomous docking, contributing to the feasibility of on-orbit refueling and sustainable satellite operations. The current study focuses exclusively on the close-range docking maneuver and does not address post-docking liquid transfer dynamics. Future work will extend the methodology to incorporate such effects, as well as environmental torques at orbital altitude.

CRediT authorship contribution statement

Mahya Ramezani: Writing – review & editing, Writing – original draft, Investigation, Conceptualization, Formal analysis, Methodology, Validation, Visualization. **M. Amin Alandihallaj:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization, Data curation, Formal analysis, Software. **Barış Can Yalçın:** Data curation, Writing – review & editing. **Miguel Angel Olivares Mendez:** Supervision. **Andreas M. Hein:** Supervision.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT-4o to check grammar. After utilizing this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the final published version of the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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